DC jack overvoltage protection, after D1 (D1 blocks reverse polarity).
U7 switches off above 1.2 V x (1 + R20/R21) = 5.9 V typ (5.7–6.1 V),
so a 9–24 V supply on J1 just doesn't power the board.

Changes for SMD5 – SMD6:

New accessory JST header (in prep for cable to payphone coin relay board.
New i2c gpio chip to handle line2 SHK/RM/FR/PD so we can run uSD AND line 2 AND 4 GPIOs
4 GPIO header now becomes a JST with power & gnd.
3.6V or 3.7V regulator for bluetooth module instead of 3.3V. Also improves headroom for 6V supply?

Fixes:
Major bug: Cheap ESP32 wide footprint had one of the 5V pins shorted to ground. Fixed in this release.
move passives in ag1171 footprint area for ag1171 real module… V5 touches caps.
bigger esp32 holes! they're a little tight

TBD:
Do we want to do SMT 470uF caps? (add pads) – not now.

U7
WS3222D-8/TR
VDC_IN
C28 1uF 50V
D13 SMAJ22A
R20 39k
R21 10k
IN
OUT
OVLO
GND
+5V

Accessory Header
Accessory
J11
I2C_SCL
I2C_SDA
+3.3V

U5
XC6206P332MR
+5V
+3.6V_BM
C26 1uF
C25 1uF
VI
VO

J1
Barrel_Jack_PTH_Slot
D1
SS34
VDC_IN

Bluetooth BM64 header
J2 PINHD_1x8_Male
+3.6V_BM
pin 5&6 matching cap mic inputs
C23 1uF
C24 1uF
C22 1uF
1k
R16
LineOVin
LineOVout
UART to bluetooth board
MFB
R17 100k

U1
ESP32-C6-DevKitC-1
3V3
EN/RESET
GPIO4/MTMS
GPIO5/MTDI
GPIO6
GPIO7
GPIO0
GPIO1
GPIO8/BOOT
GPIO10
GPIO11
GPIO2
GPIO3
5V
GND
NC
GPIO16/U0TXD
GPIO17/U0RXD
GPIO15/JTAG
GPIO23
GPIO22
GPIO21
GPIO20
GPIO19
GPIO18
GPIO9/BOOT
GPIO13/USB_D+
GPIO12/USB_D–
Line0_F/R
Line0_RM
Line0_SHK
Line0_PD
I2S_DIN
I2S_LRCK
I2S_DOUT
I2S_BCLK
I2C_SCL
I2C_SDA
SD_CS
SD_MOSI
SD_CLK
SD_CMD
LED on GPIO8
I2S_MCLK
GPIO12
SD_CD
GPIO13
JP3
JP2
Remove jumpers for uSD use
PINHD_1x2_Male
J4
J5
Pull up to VCC in SW for switches/IO

Caps as close to their respective gnd returns as possible
C7 10uF
C8 100nF
C5 100nF
C6 100nF
R5 10R
C9 100nF

U2
ES8388
MCLK
SCLK
LRCK
DSDIN
ASDOUT
CDATA
CCLK
CE
VREF
ADCVREF
VMID
DVDD
PVDD
AVDD
HPVDD
DGND
AGND
HPGND
EPAD
LIN1
RIN1
LIN2
RIN2
LOUT1
ROUT1
LOUT2
ROUT2
CH0_Spk
CH1_Spk
CH 2 out is a copy of CH 1
C20 100nF
C21 100nF
R8 10k
R10 10k
CH0_Mic
R9 10k
R11 10k
CH1_Mic
5V AG1170 needs 3.3V signal to ESP32

I2C pull-ups
R3 4.7k
R4 4.7k
CE Gnd = i2c addr 0x10

C2 1uF
C3 1uF
C4 1uF

J7
PINHD_1x3_Male
R6 47k
R7 47k
C10 47uF
C11 47uF
Pin 1: L/CH0
Pin 2: R/CH1
Optional Headphone header

R12 100k
R13 10k
Line1_SHK
Line1_RM
Line1_F/R
Zener BZT03C82 OR P6KE82
D8 SM4007PL
D4 SM4007PL
D5 SM4007PL
D9 SM4007PL
J8
RJ12
C14 100nF
C15 10nF
CH1_Mic
CH1_Spk
U3
AG1171
Line 1: Right
D2
1N4148
Line1_PD
Diode
1N4148 or BAS16
C18 100nF
C16 470uF 16V

R14 100k
R15 10k
Line0_SHK
Line0_RM
Line0_F/R
D10SM4007PL
D6 SM4007PL
D12 D_Zener P6KE82
D7 SM4007PL
D11 SM4007PL
J9
RJ12
U4
AG1171
Line 0: Left
D3
1N4148
Line0_PD
Diode
1N4148 or BAS16
C13 100nF
C12 10nF
CH0_Mic
CH0_Spk
C19 100nF
C17 470uF 16V

H1 MountingHole_M3
H2 MountingHole_M3
H3 MountingHole_M3
H4 MountingHole_M3

CF Card
Card1
TF-01A
R1 10k
R2 10k
SD_MISO
SD_CS
SD_MOSI
SD_CLK
SD_CD
DAT0(DO)
DAT1(RSV)
DAT2(RSV)
CDDAT3(CS)
CMD(DI)
CLK(SCLK)
VDD
VSS
CD
C1 100nF

U6
TCA9534APWR
C27 100nF
I2C_SCL
I2C_SDA
INT
A0
A1
A2
P0
P1
P2
P3
P4
P5
P6
P7
Line1_F/R
Line1_RM
Line1_SHK
Line1_PD
EXP_P4
EXP_P5
EXP_P6
EXP_P7
R18 10k
R19 10k
J10
GPIO Expansion

Line 2 + microSD together: U6 (TCA9534A, I2C 0x38) drives line 2's SLIC.
Leave J5 open when U6 is fitted (J5 = line 2 on GPIO18–21 instead of the card).
–INT unused: firmware polls SHK. R18/R19 keep F/R and RM low until U6 is set up.
J10: spare expander pins P4–P7.

Components only – manual routing

Sheet: /
File: Analog Redial v6.kicad_sch

Title: Rotary Phone SIP Gateway

Size: A3	Date:	Rev: 0.5
KiCad E.D.A. 10.0.3		Id: 1/1